EXERCISES 3.1

Answers to selected odd-numbered problems begin on page ANS-10.

3.1.1 Limits

In Problems 1–8, use Theorem 3.1.1 and the properties of real limits on page 103 to compute the given complex limit.

1. $\lim_{z\to 2i}\left(z^2-\bar{z}\right)$

2. $\lim_{z\to 1+i}\dfrac{z-\bar{z}}{z+\bar{z}}$

3. $\lim_{z\to 1-i}\left(|z|^2-i\bar{z}\right)$

4. $\lim_{z\to 3i}\dfrac{\text{Im}\left(z^2\right)}{z+\text{Re}\,(z)}$

5. $\lim_{z\to \pi i} e^z$

6. $\lim_{z\to 0}\dfrac{z^2+\bar{z}^2}{\text{Re}(z)+\text{Im}(z)}$

7. $\lim_{z\to 0}\dfrac{e^z-e^{\bar{z}}}{\text{Im}(z)}$

8. $\lim_{z\to 1+i}\left(\log_e\left|x^2+y^2\right|+i\arctan\dfrac{y}{x}\right)$

In Problems 9–16, use Theorem 3.1.2 and the basic limits (15) and (16) to compute the given complex limit.

9. $\lim_{z\to 2-i}\left(z^2-z\right)$

10. $\lim_{z\to i}\left(z^5-z^2+z\right)$

11. $\lim_{z\to e^{i\pi/4}}\left(z+\dfrac{1}{z}\right)$

12. $\lim_{z\to 1+i}\dfrac{z^2+1}{z^2-1}$

13. $\lim_{z\to -i}\dfrac{z^4-1}{z+i}$

14. $\lim_{z\to 2+i}\dfrac{z^2-(2+i)^2}{z-(2+i)}$

15. $\lim_{z\to z_0}\dfrac{(az+b)-(az_0+b)}{z-z_0}$

16. $\lim_{z\to -3+i\sqrt{2}}\dfrac{z+3-i\sqrt{2}}{z^2+6z+11}$

17. Consider the limit $\lim_{z\to 0}\dfrac{\text{Re}(z)}{\text{Im}(z)}$.

(**a**) What value does the limit approach as z approaches 0 along the line $y=x$?

(**b**) What value does the limit approach as z approaches 0 along the imaginary axis?

(**c**) Based on your answers for (a) and (b), what can you say about $\lim_{z\to 0}\dfrac{\text{Re}(z)}{\text{Im}(z)}$?

18. Consider the limit $\lim_{z\to i}(|z|+i\,\text{Arg}\,(iz))$.

(**a**) What value does the limit approach as z approaches i along the unit circle $|z|=1$ in the first quadrant?

(**b**) What value does the limit approach as z approaches i along the unit circle $|z|=1$ in the second quadrant?

(**c**) Based on your answers for (a) and (b), what can you say about $\lim_{z\to i}(|z|+i\,\text{Arg}\,(iz))$?

19. Consider the limit $\lim_{z\to 0}\left(\dfrac{z}{\bar{z}}\right)^2$.

(**a**) What value does the limit approach as z approaches 0 along the real axis?

(**b**) What value does the limit approach as z approaches 0 along the imaginary axis?

(**c**) Do the answers from (a) and (b) imply that $\lim_{z\to 0}\left(\dfrac{z}{\bar{z}}\right)^2$ exists? Explain.

(**d**) What value does the limit approach as z approaches 0 along the line $y=x$?

(**e**) What can you say about $\lim_{z\to 0}\left(\dfrac{z}{\bar{z}}\right)^2$?

20. Consider the limit $\lim\limits_{z\to 0}\left(\dfrac{2y^2}{x^2}-\dfrac{x^2-y^2}{y^2}i\right)$.

(**a**) What value does the limit approach as z approaches 0 along the line $y=x$?

(**b**) What value does the limit approach as z approaches 0 along the line $y=-x$?

(**c**) Do the answers from (a) and (b) imply that $\lim\limits_{z\to 0}\left(\dfrac{2y^2}{x^2}-\dfrac{x^2-y^2}{y^2}i\right)$ exists? Explain.

(**d**) What value does the limit approach as z approaches 0 along the line $y=2x$?

(**e**) What can you say about $\lim\limits_{z\to 0}\left(\dfrac{2y^2}{x^2}-\dfrac{x^2-y^2}{y^2}i\right)$?

Problems 21–26 involve concepts of infinite limits and limits at infinity discussed in (*i*) of the *Remarks*. In Problems 21–26, use (21) or (22), Theorem 3.1.2, and the basic limits (15) and (16) to compute the given complex limit.

21. $\lim\limits_{z\to\infty}\dfrac{z^2+iz-2}{(1+2i)z^2}$

22. $\lim\limits_{z\to\infty}\dfrac{iz+1}{2z-i}$

23. $\lim\limits_{z\to i}\dfrac{z^2-1}{z^2+1}$

24. $\lim\limits_{z\to -i/2}\dfrac{(1-i)z+i}{2z+i}$

25. $\lim\limits_{z\to\infty}\dfrac{z^2-(2+3i)z+1}{iz-3}$

26. $\lim\limits_{z\to i}\dfrac{z^2+1}{z^2+z+1-i}$

3.1.2 Continuity

In Problems 27–34, show that the function f is continuous at the given point.

27. $f(z)=z^2-iz+3-2i;\ z_0=2-i$

28. $f(z)=z^3-\dfrac{1}{z};\ z_0=3i$

29. $f(z)=\dfrac{z^3}{z^3+3z^2+z};\ z_0=i$

30. $f(z)=\dfrac{z-3i}{z^2+2z-1};\ z_0=1+i$

31. $f(z)=\begin{cases}\dfrac{z^3-1}{z-1}, & |z|\neq 1\\ 3, & |z|=1\end{cases};\ z_0=1$

32. $f(z)=\begin{cases}\dfrac{z^3-1}{z^2+z+1}, & |z|\neq 1\\ \dfrac{-1+i\sqrt{3}}{2}, & |z|=1\end{cases};\ z_0=\dfrac{-1+i\sqrt{3}}{2}$

33. $f(z)=\bar{z}-3\,\mathrm{Re}(z)+i;\ z_0=3-2i$

34. $f(z)=\dfrac{\mathrm{Re}(z)}{z+iz}-2z^2;\ z_0=e^{i\pi/4}$

In Problems 35–40, show that the function f is discontinuous at the given point.

35. $f(z)=\dfrac{z^2+1}{z+i};\ z_0=-i$

36. $f(z)=\dfrac{1}{|z|-1};\ z_0=i$

37. $f(z)=\mathrm{Arg}(z);\ z=-1$

38. $f(z)=\mathrm{Arg}(iz);\ z_0=i$

39. $f(z) = \begin{cases} \dfrac{z^3-1}{z-1}, & |z| \neq 1 \\ 3, & |z| = 1 \end{cases}$; $z_0 = i$

40. $f(z) = \begin{cases} \dfrac{z}{|z|}, & z \neq 0 \\ 1, & z = 0 \end{cases}$; $z_0 = 0$

In Problems 41–44, use Theorem 3.1.3 to determine the largest region in the complex plane on which the function f is continuous.

41. $f(z) = \text{Re}(z)\,\text{Im}(z)$

42. $f(z) = \bar{z}$

43. $f(z) = \dfrac{z-1}{z\bar{z}-4}$

44. $f(z) = \dfrac{z^2}{(|z|-1)\,\text{Im}(z)}$

Focus on Concepts

45. Use Theorem 3.1.1 to prove:

(**a**) $\lim\limits_{z\to z_0} c = c$, where c is a constant.

(**b**) $\lim\limits_{z\to z_0} z = z_0$.

46. Use Theorem 3.1.1 to show that $\lim\limits_{z\to z_0} \bar{z} = \bar{z}_0$.

47. Use Theorem 3.1.2 and Problem 46 to show that

(**a**) $\lim\limits_{z\to z_0} \text{Re}(z) = \text{Re}(z_0)$.

(**b**) $\lim\limits_{z\to z_0} \text{Im}(z) = \text{Im}(z_0)$.

(**c**) $\lim\limits_{z\to z_0} |z| = |z_0|$.

48. Use Theorem 3.1.1 to prove part (*ii*) of Theorem 3.1.2.

49. The following is an epsilon-delta proof that $\lim\limits_{z\to z_0} z = z_0$. Fill in the missing parts.

Proof By Definition 3.1.1, $\lim\limits_{z\to z_0} z = z_0$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|_____| < \varepsilon$ whenever $0 < |_____| < \delta$. Setting $\delta =$ _____ will ensure that the previous statement is true.

50. The following is an epsilon-delta proof that $\lim\limits_{z\to z_0} \bar{z} = \bar{z}_0$. Provide the missing justifications in the proof.

Proof By Definition 3.1.1, $\lim\limits_{z\to z_0} \bar{z} = \bar{z}_0$ if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|_____| < \varepsilon$ whenever $0 < |_____| < \delta$. By properties of complex modulus and conjugation, $|z - z_0| = \left|\overline{z - z_0}\right| = |_____|$. Therefore, if $0 < |z - z_0| < \delta$ and $\delta =$ _____, then $|\bar{z} - \bar{z}_0| < \varepsilon$.

51. In this problem we will develop an epsilon-delta proof that $\lim\limits_{z\to 1+i} ((1-i)\,z + 2i) = 2 + 2i$.

(**a**) Write down the epsilon-delta definition (Definition 3.1.1) of $\lim\limits_{z\to 1+i} [(1-i)\,z + 2i] = 2 + 2i$.

(**b**) Factor out $(1-i)$ from the inequality involving ε (from part (a)) and simplify. Now rewrite this inequality in the form $|z - (1+i)| <$ _____.

(**c**) Based on your work from part (b), what should δ be set equal to?

(**d**) Write an epsilon-delta proof that $\lim\limits_{z\to 1+i} [(1-i)\,z + 2i] = 2 + 2i$.

52. In this problem we will develop an epsilon-delta proof that $\lim\limits_{z\to 2i} \dfrac{2z^2 - 3iz + 2}{z - 2i} = 5i$.

(**a**) Write down the epsilon-delta definition (Definition 3.1.1) of $\lim\limits_{z\to 2i} \dfrac{2z^2 - 3iz + 2}{z - 2i} = 5i$.

(**b**) Simplify the inequality involving ε (from part (a)), then rewrite this inequality in the form $|z - 2i| <$ _____.

(**c**) Based on your work from part (b), what should δ be set equal to?

(**d**) Write an epsilon-delta proof that $\lim\limits_{z\to 2i}\dfrac{2z^2-3iz+2}{z-2i}=5i$.

53. (**a**) Is it true that $\lim\limits_{z\to z_0}\overline{f(z)}=\lim\limits_{z\to z_0}f(\bar{z})$ for any complex function f? If so, then give a brief justification; if not, then find a counterexample.

(**b**) If $f(z)$ is a continuous function at z_0, then is it true that $\overline{f(z)}$ is continuous at z_0?

54. If f is a function for which $\lim\limits_{x\to 0}f(x+i0)=0$ and $\lim\limits_{y\to 0}f(0+iy)=0$, then can you conclude that $\lim\limits_{z\to 0}f(z)=0$? Explain.

55. (**a**) Prove that the function $f(z)=\text{Arg}(z)$ is discontinuous at every point on the negative real axis.

(**b**) Prove that the function f_1 defined by

$$f_1(z)=\theta,\ -\pi<\theta<\pi$$

is a branch of the multiple-valued function $F(z)=\arg(z)$. [*Hint*: See Example 8.]

56. Consider the multiple-valued function $F(z)=z^{1/3}$ that assigns to z the set of three cube roots of z. Explicitly define three distinct branches f_1, f_2, and f_3 of F, all of which have the nonnegative real axis as a branch cut.

57. Consider the multiple-valued function $F(z)=(z-1+i)^{1/2}$.

(**a**) What is the branch point of F? Explain.

(**b**) Explicitly define two distinct branches of f_1 and f_2 of F. In each case, state the branch cut.

58. Consider the multiple-valued function $F(z)=\left(z^2+1\right)^{1/2}$. What are the branch points (there are two of them) of F? Explain.

Computer Lab Assignments

Reread part (*ii*) of the *Remarks* at the end of Section 3.1. In Problems 59–62, use a CAS to show that the given function is not continuous inside the unit circle by plotting the image of the given continuous parametric curve. (Be careful, *Mathematica* and *Maple* plots can sometimes be misleading.)

59. $f(z)=z+\text{Arg}(z),\ z(t)=-\frac{1}{2}+\frac{1}{2}\sqrt{3}it,\ -1\le t\le 1$

60. $f(z)=\sqrt[4]{r}e^{i\theta/4},\ \theta=\text{Arg}(z),\ z(t)=-\frac{1}{2}+\frac{1}{2}\sqrt{3}it,\ -1\le t\le 1$

61. $f(z)=\sqrt{r}e^{i\theta/2},\ \theta=\text{Arg}(z),\ z(t)=-\frac{1}{2}+\frac{1}{4}e^{it},\ 0\le t\le 2\pi$

62. $f(z)=|z-1|\text{Arg}(-z)+i\,\text{Arg}(iz),\ z(t)=\frac{1}{2}e^{it},\ 0\le t\le 2\pi$

3.2 Differentiability and Analyticity

INTRODUCTION The calculus of complex functions deals with the usual concepts of derivatives and integrals of these functions. In this section we shall give the limit definition of the derivative of a complex function $f(z)$. Although many of the concepts in this section will seem familiar to you, such as the product, quotient, and chain rules of differentiation, there are important differences between this material and the calculus of real functions $f(x)$. As the subsequent chapters of this text unfold, you will see that except for familiarity of names and definitions, there is little similarity between the *interpretations* of quantities such as $f'(x)$ and $f'(z)$.